Shubham Pandey

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Generative AI Engineer (2.5+ years) building LLM systems, including production RAG, vision-language document extraction, LLM evaluation and context engineering, on a strong Python, data-engineering and AWS foundation.

Professional Experience

Impetus Technologies Oct 2025 – Present
Generative AI Engineer *Remote, India*

- **Finance Document Intelligence Platform (Pilot)** | *Python, FastAPI, Qwen VLM, Qdrant, Celery, Langfuse*
  - Engineered REST APIs and **LLM extraction pipelines** on **Qwen VLM** with semantic chunking and embeddings, turning **5+ unstructured formats** (PDF, images, Excel, PPTX) into structured data.
  - Designed an **automated accuracy-evaluation module** comparing each source PDF with its Markdown extraction to flag gaps and score accuracy; with QA feedback loops, raised accuracy from **60% to 85%+** on a **90+ document** QA set.
  - Implemented Qdrant semantic search for retrieval and Langfuse tracing for end-to-end **LLM observability**, latency and cost monitoring, with async processing on Celery and Redis.
- **RAG-Based Sales Intelligence Platform** | *Python, AWS Bedrock, FastAPI, Kubernetes, PostgreSQL*
  - Shipped a **production RAG pipeline** on AWS Bedrock delivering deal intelligence, win-probability analysis and objection handling to **200+ sales users**.
  - Engineered **deal-aware LLM response caching** in PostgreSQL, cutting repeat-query latency from **35s to 2s** and avoiding **3–5 repeat LLM calls per user**; added Redis per-user rate limiting to cap LLM cost.
  - Automated daily AI nudges emailing prioritized insights and next actions to sales reps across **5,000+ deals**.
- **Context Engineering PoCs:** added semantic definitions, metrics and a Turtle (RDF) ontology for **Databricks Genie** NL analytics; built most of an end-to-end **AWS Context Ontology Accelerator** PoC.

Tata Consultancy Services (TCS) Jan 2024 – Oct 2025
Python Developer | ETL & Data Engineering *Pune, India*

- Built a RAG-based **production support assistant** over the team knowledge base to speed up issue resolution.
- Optimized business-critical ETL pipelines in Python and SQL, cutting execution time by **50%**.
- Tuned PostgreSQL queries and indexes, cutting average response times by over **60%**.

Projects

Samajh – AI Diagnostic Testing & Exam Prep | *Python, FastAPI, PostgreSQL, Gemini, Render* | Live

- Built a bilingual (Hindi/English) testing platform where each wrong option maps to a named **misconception**, giving per-student "why" reports and class re-teach plans, plus full UP Super TET and PCS mocks.
- Designed an **AI question pipeline**: Gemini drafts against JSON schemas, a second model **solves blind** to catch wrong keys, a teacher approves; **500+ questions**, **2 errors** in a 65-question hand check.
- Added **LLM rubric grading** of free-text answers that defers to the teacher instead of guessing when the model fails, a model fallback chain for quota limits, CSP/CSRF/rate-limit protection and **41 tests**.

Live AI Hindi Cricket Commentary | *Python, FastAPI, WebSockets, Gemini, TTS, Docker* | GitHub

- Built a **real-time pipeline** turning ball-by-ball data into spoken Hindi commentary with **4 AI personas**.
- Added **LLM guardrails** (JSON-schema outputs, a **3.5s timeout**, a fact checker against match state, template fallback), resilient multi-provider TTS, a replay harness and regression tests from live-match bugs.

Education

United College of Engineering & Research 2019 – 2023
Bachelor of Technology, Computer Science *Prayagraj, India*

- TCS CodeVita Season 10: Global Rank **3,370**, College Rank **3rd**

Technical Skills

LLM & GenAI: RAG, Context Engineering, LLM Evaluation, LLM-as-judge, Prompt Engineering, Structured Outputs, Ontologies
Platforms & Models: AWS Bedrock, Databricks Genie, Google Gemini, Qwen VLM, Langfuse, Qdrant, FAISS
Languages & Backend: Python, SQL, FastAPI, Django REST Framework, WebSockets, Celery, PySpark
Data & Infra: PostgreSQL, Redis, Docker, Kubernetes, Airflow, AWS (S3, RDS, Redshift, Lambda, Glue, Athena), Git